

THE STRUCTURE OF MATHEMATICAL REALITY

Phase III — Chapter 10

Dynamics and Approximation: The Structure of Change

Assignment

Self-Study Curriculum

Choose a system from your own work that changes over time (a population, a reaction, a physiological signal, a culture). Write a one-page structural analysis:

1. What is the *state* (the minimal variables needed to determine the future)? Justify the minimality.
2. Write down, or describe qualitatively, the rule of evolution. Where are the feedback loops, and what are their signs?
3. Identify the equilibria. For at least one, linearize (Jacobian, or just the sign of the local slope) and classify its stability.
4. Is there any regime where linearization fails, a switch, an oscillation, a threshold, a tipping point? Locate it and say which nonlinear phenomenon it is.
5. State one prediction the *dynamical* view makes that a purely statistical model of the same data would miss.